Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at /home/guest/ENV872/EDE_Bella
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
getwd()
```

```
## [1] "/home/guest/ENV872/EDE_Bella"
```

```r
#import the two datasets
ntl_peterpaul <- read_csv(here("Data", "Processed_KEY", "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Pr
```

```
## Rows: 23008 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (1): lakename
## dbl  (13): year4, daynum, month, depth, temperature_C, dissolvedOxygen, irra...
## date  (1): sampledate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
niwot_litter  <- read_csv(here("Data", "Processed_KEY", "NEON_NIWO_Litter_mass_trap_Processed.csv"))
```

```
## Rows: 1692 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (7): plotID, trapID, functionalGroup, qaDryMass, nlcdClass, plotType, g...
## dbl  (5): dryMass, subplotID, decimalLatitude, decimalLongitude, elevation
## date (1): collectDate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#2
#check the type of date in the two datasets
class(ntl_peterpaul$sampledate)
```

```
## [1] "Date"
```

```
class(niwot_litter$collectDate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
# set background color & put title on top in the middle & place legend on top
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        panel.background = element_rect(fill = "gray95", color = NA),
        plot.background = element_rect(fill = "white", color = NA),
        plot.title = element_text(
          hjust = 0.5,
          face = "bold"),
        legend.position = "top")
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

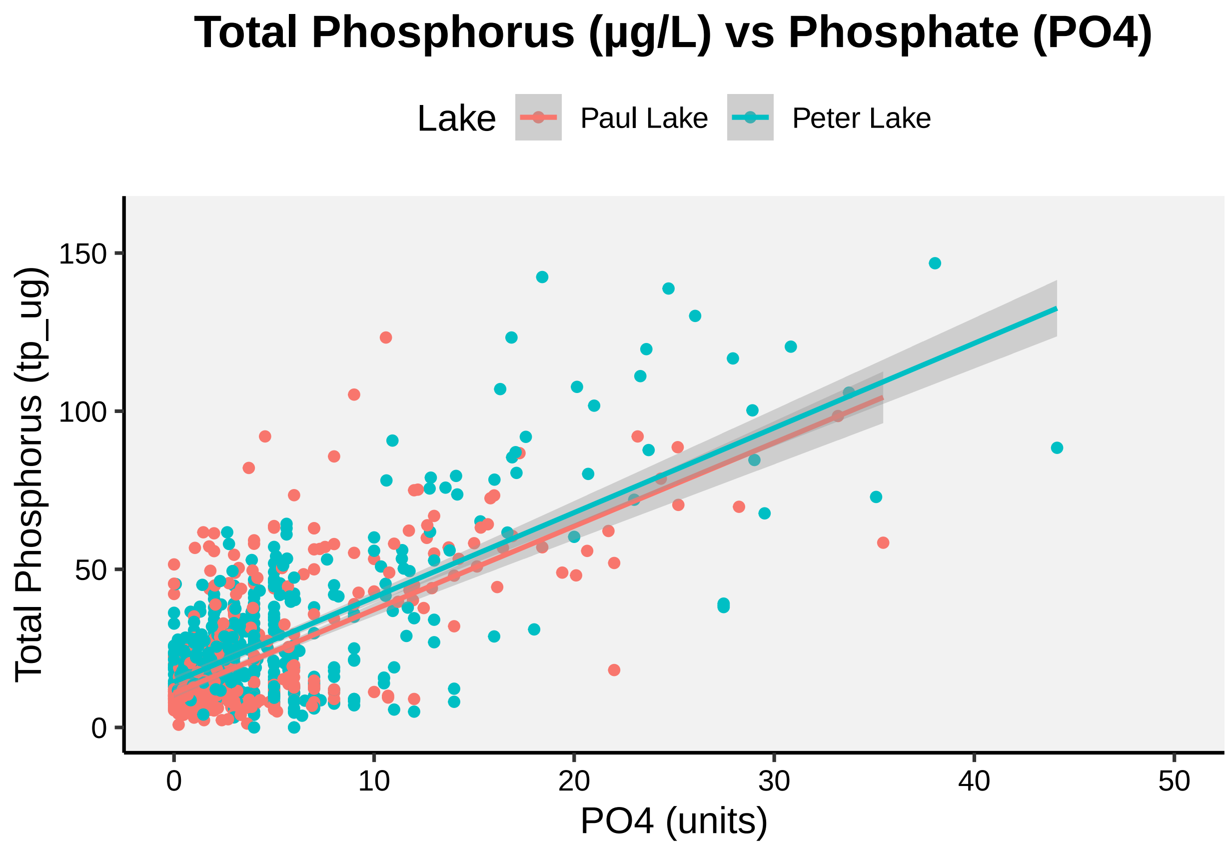
4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
library(ggplot2)
ggplot(ntl_peterpaul, aes(x=po4, y=tp_ug, color=lakename)) +
         geom_point()+
         geom_smooth(method = "lm", se = TRUE, linewidth = 0.9)+
        xlim(0, 50) +
        ylim(0, 160) +
         labs(
    title = "Total Phosphorus (µg/L) vs Phosphate (PO4)",
    x = "PO4 (units)",
    y = "Total Phosphorus (tp_ug)",
    color = "Lake")+
    mytheme
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 21948 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21948 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
#temperature
temperature <- ntl_peterpaul %>%
  #filter(depth == 0) %>%
  mutate(month = factor(month, levels = 1:12, labels = month.abb)) %>%
  ggplot(aes(x = month, y = temperature_C, color = lakename)) +
  geom_boxplot(outlier.alpha = 0.4, width=0.7) +
  scale_x_discrete(drop = FALSE) +
  labs(title = "A: Temperature", y = "Temperature (°C)", fill = "Lake") +
  mytheme+
```

```r
  theme(axis.title.x = element_blank(),
        legend.position = "none")

#TP
tp <- ntl_peterpaul %>%
  #filter(depth == 0) %>%
  mutate(month = factor(month, levels = 1:12, labels = month.abb)) %>%
  ggplot(aes(x = month, y = tp_ug, color = lakename)) +
  geom_boxplot(outlier.alpha = 0.4, width=0.7) +
  scale_x_discrete(drop = FALSE) +
  labs(title = "B: Total Phosphorus (TP)", y = "TP (mg/L)", fill = "Lake") +
  mytheme+
  theme(axis.title.x = element_blank(),
        legend.position = "none")
#TN
tn <- ntl_peterpaul %>%
  #filter(depth == 0) %>%
  mutate(month = factor(month, levels = 1:12, labels = month.abb)) %>%
  ggplot(aes(x = month, y = tn_ug, color = lakename)) +
  geom_boxplot(outlier.alpha = 0.4, width=0.7) +
  scale_x_discrete(drop = FALSE) +
  labs(title = "C: Total Nitrogen (TN)", y = "TN (mg/L)", fill = "Lake") +
  mytheme+
  theme(axis.title.x = element_blank(),
        legend.position = "bottom")


shared_legend <- get_legend(tn + theme(legend.position = "bottom",
                                        legend.direction = "horizontal",
                                        legend.title = element_text(face = "bold"),
                                        legend.text = element_text(size = 11)))
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```

```r
# remove legends from single panels for clean stacking
p_temp_noleg <- temperature + theme(legend.position = "none")
p_tp_noleg   <- tp    + theme(legend.position = "none")
p_tn_noleg   <- tn    + theme(legend.position = "none")

# align the three plots' widths so boxes line up exactly
aligned <- align_plots(p_temp_noleg, p_tp_noleg, p_tn_noleg, align = "v", axis = "lr")
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

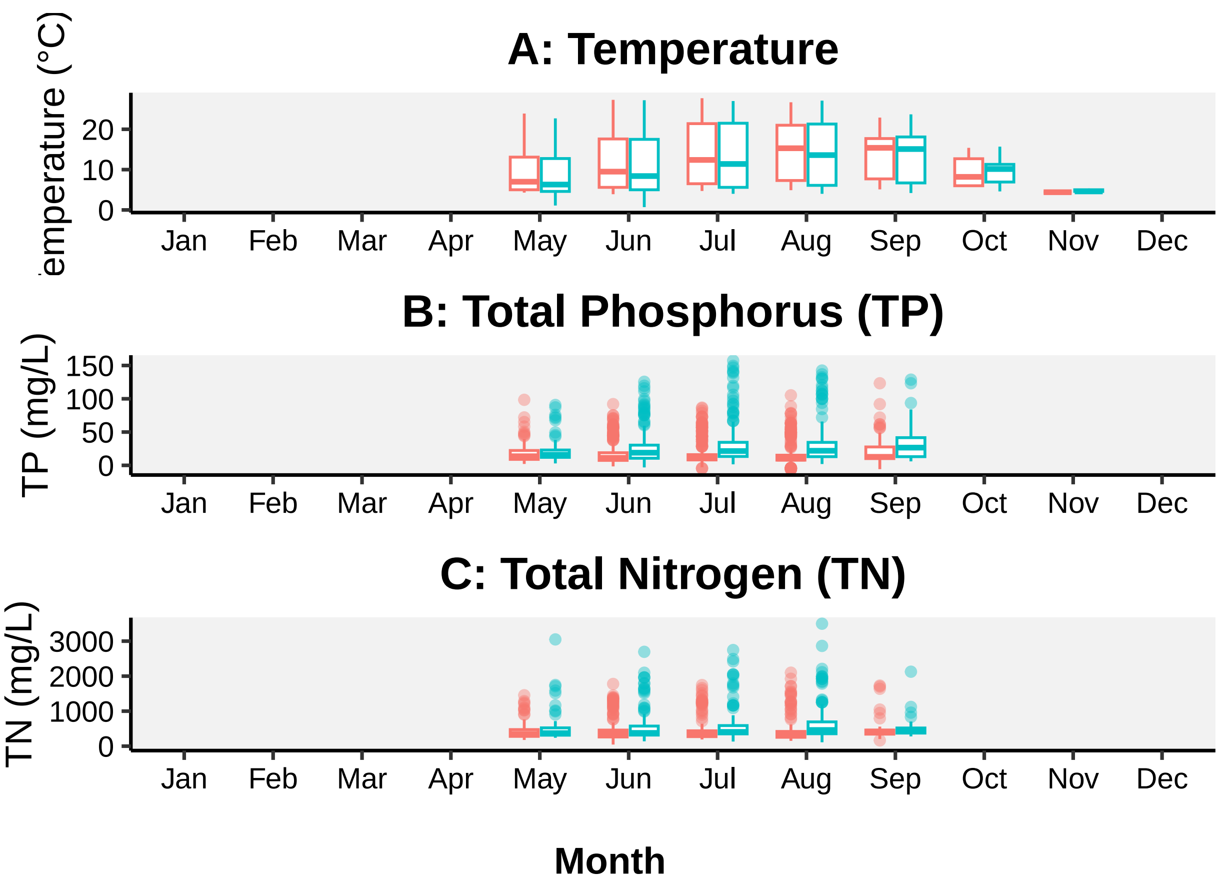
```
p1 <- aligned[[1]]; p2 <- aligned[[2]]; p3 <- aligned[[3]]

# stack vertically with a little more room for the bottom panel
combined <- plot_grid(p1, p2, p3, ncol = 1, align = "v", axis = "lr", rel_heights = c(1,1,1.05))

# put legend below and add shared x-axis label
plots_with_legend <- plot_grid(combined, shared_legend, ncol = 1, rel_heights = c(1, 0.08))

final <- ggdraw(plots_with_legend) +
  draw_label("Month", x = 0.5, y = 0.02, fontface = "bold", size = 14)

# display
print(final)
```

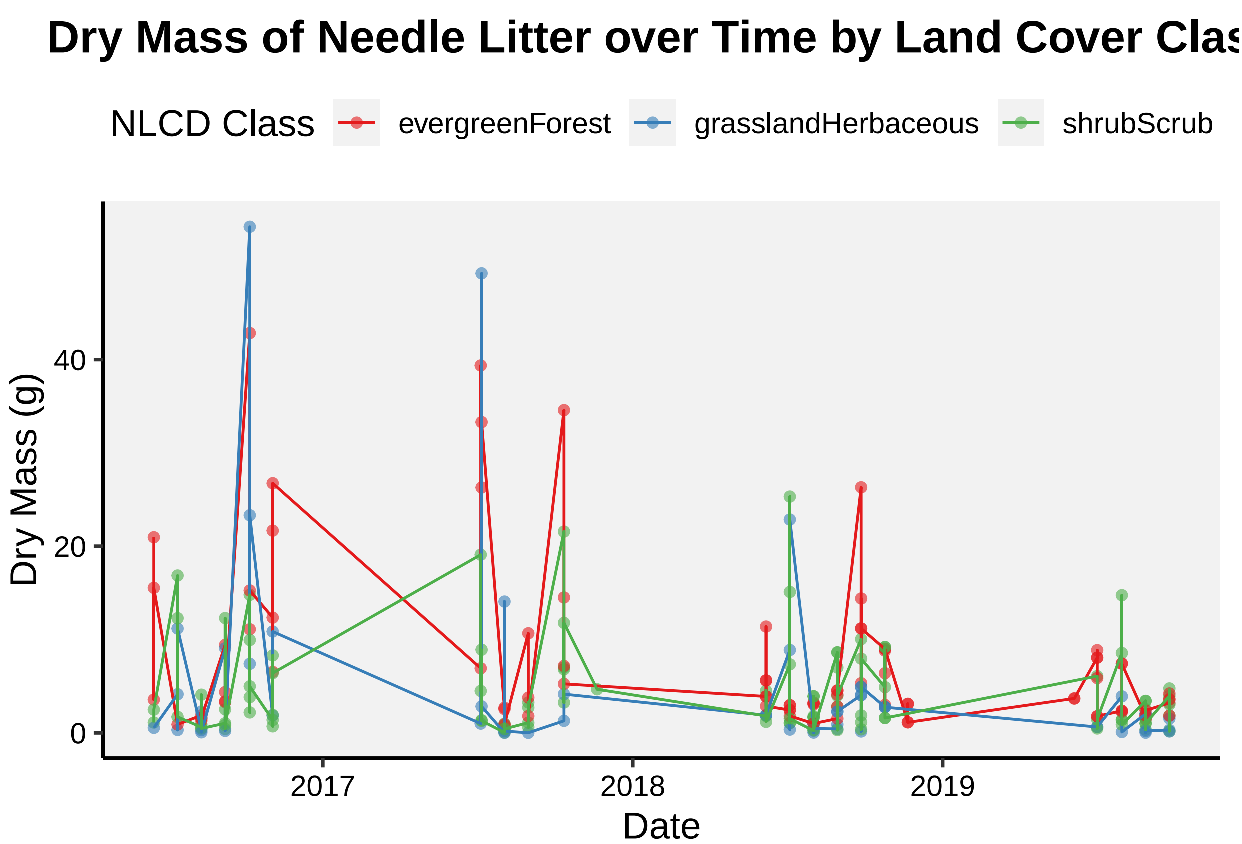


Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer:The temperature trend for the two lakes are very similar, and shows a seasonal trend: it increases from May through summer, and drops again by October. After that, no data, which might indicate that they all got frozen. For TP level, Paul Lake has slightly higher median TP values than Peter Lake across months. For both lakes, the TP value is lowest in May and June, while increased during July and August. For TN level, Paul Lake again has slightly higher TN concentrations. For the two lakes, the TN concentration is quite stable throughout the months.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

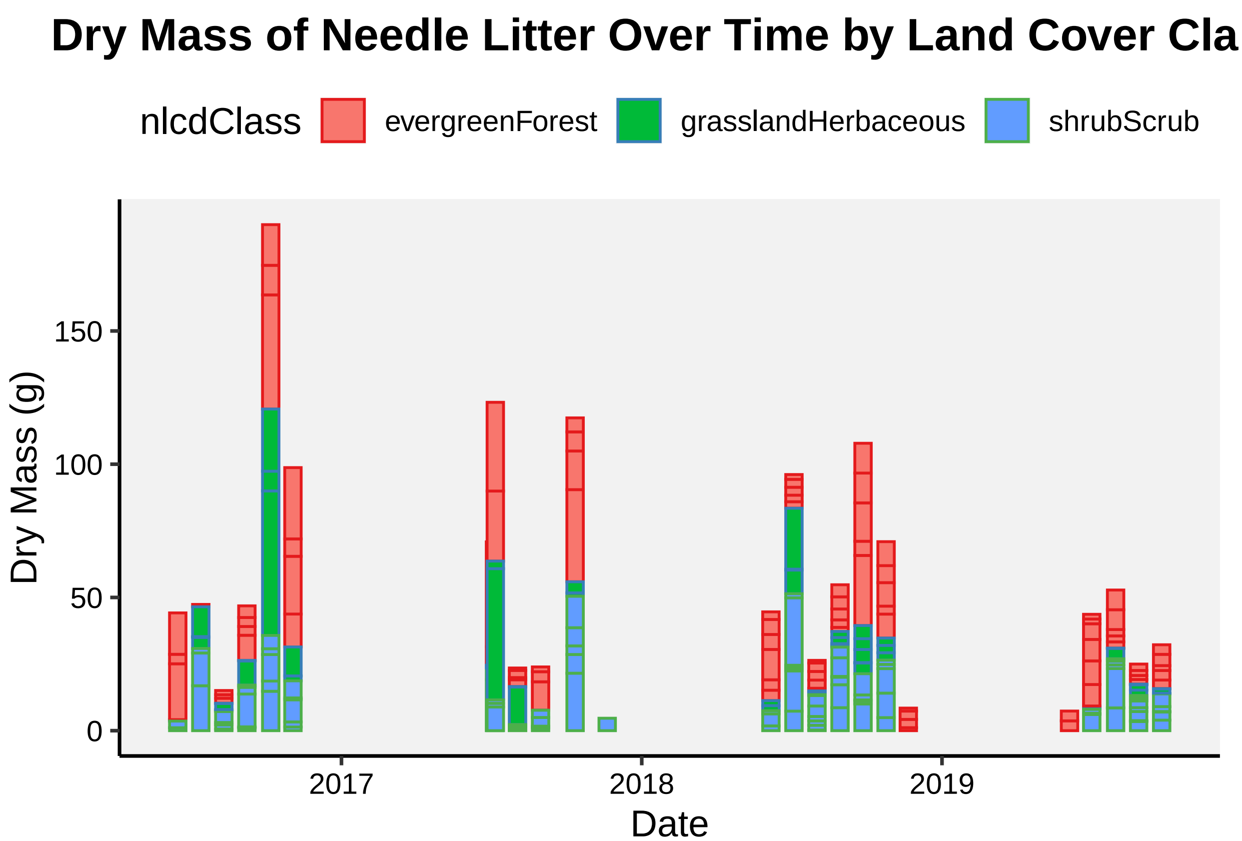
```
#6
#filter Needles out
Needles <- niwot_litter %>%
  filter(functionalGroup == "Needles")
#plot 1
ggplot(Needles, aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(alpha = 0.6) +
  geom_line() +
  labs(
    title = "Dry Mass of Needle Litter over Time by Land Cover Class",
    x = "Date",
    y = "Dry Mass (g)",
    color = "NLCD Class") +
  scale_color_brewer(palette = "Set1") +
  mytheme
```



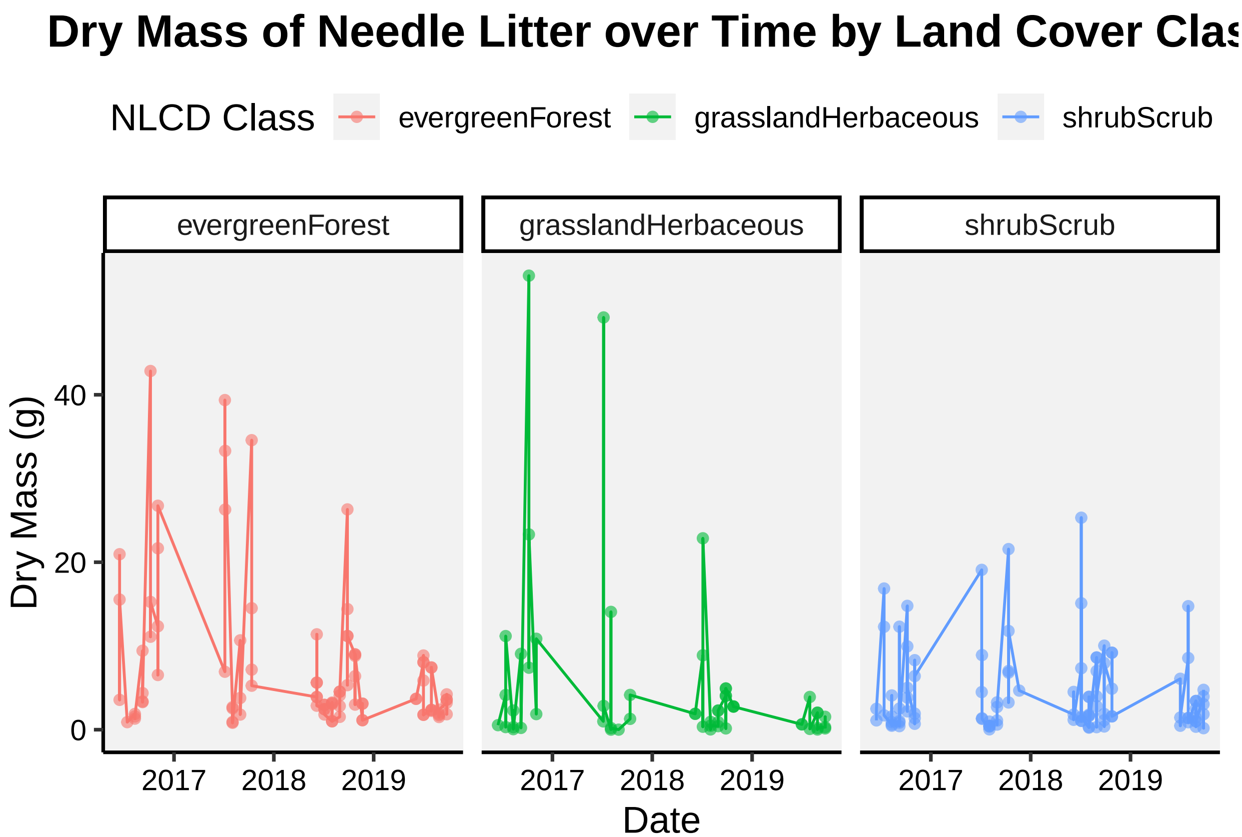

```
#plot 2 -- Bar since line is so messy
ggplot(Needles, aes(x = collectDate, y = dryMass, fill = nlcdClass, color = nlcdClass)) +
  geom_col(width = 20) +
```

```
  labs(
    title = "Dry Mass of Needle Litter Over Time by Land Cover Class",
    x = "Date",
    y = "Dry Mass (g)"
  ) +
  scale_color_brewer(palette = "Set1") +
  mytheme
```

```
## Warning: 'position_stack()' requires non-overlapping x intervals.
```

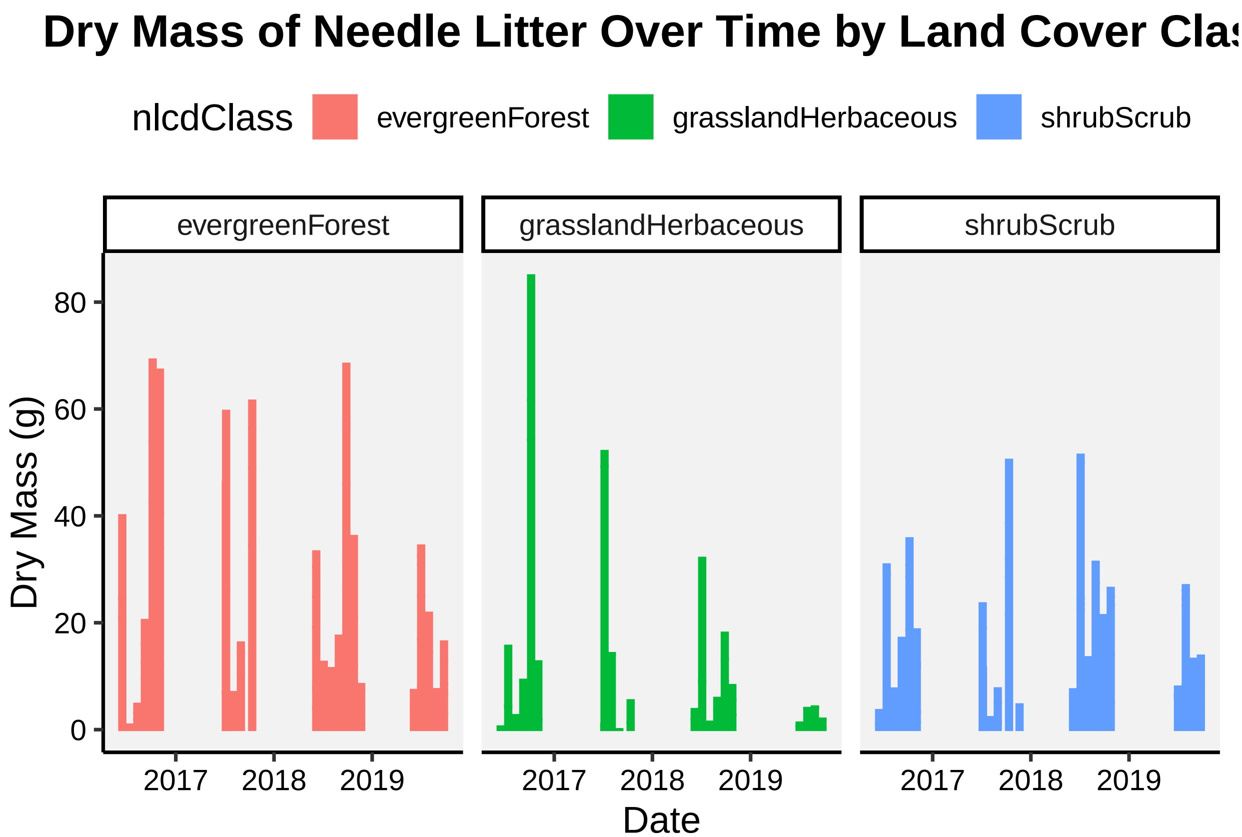



```
#7
#plot 1
ggplot(Needles, aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(alpha = 0.6) +
  geom_line() +
  facet_wrap(~ nlcdClass, ncol = 3) +
  labs(
    title = "Dry Mass of Needle Litter over Time by Land Cover Class",
    x = "Date",
    y = "Dry Mass (g)",
    color = "NLCD Class"
  ) +
  mytheme
```




```r
#plot 2
ggplot(Needles, aes(x = collectDate, y = dryMass, fill = nlcdClass, color = nlcdClass)) +
  geom_col(width = 20) +
  facet_wrap(~ nlcdClass, ncol = 3) +
  labs(
    title = "Dry Mass of Needle Litter Over Time by Land Cover Class",
    x = "Date",
    y = "Dry Mass (g)"
  ) +
  mytheme
```

```
## Warning: ‘position_stack()‘ requires non-overlapping x intervals.
## ‘position_stack()‘ requires non-overlapping x intervals.
## ‘position_stack()‘ requires non-overlapping x intervals.
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer:To me, the plots in 7 makes more sence to me and conveys the information more effectively. Since we have 3 class of NLCD, ploting with a single panel makes the data clustered together. In addition to that, the plots of 7 helps show the trend better and clearer.